

Infinite log-concavity of cycle chromatic coefficients: a complete classification

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Abstract

We classify infinite log-concavity of the full absolute chromatic coefficient sequence of every cycle: it holds exactly for lengths 3 through 11. Explicit negative iterates handle lengths 12 through 16, and a uniform third-iterate identity excludes all larger lengths. A strengthened log-concavity invariant proves all infinitely many positive iterates. The complete chain, from actual graph colorings to the if-and-only-if classification, has a Lean proof.

1 Statement and context

Let C_n be the undirected cycle with $n \geq 3$ vertices. For its chromatic polynomial $\chi_n(q)$, write $a_k = |[q^k]\chi_n(q)|$, and extend this sequence by zero to every $k \in \mathbb{Z}$ outside $0 \leq k \leq n$. For any zero-extended finite sequence define

$$(\mathcal{L}a)_k = a_k^2 - a_{k-1}a_{k+1}.$$

A sequence is infinitely log-concave if every entry of $\mathcal{L}^r a$ is nonnegative for every integer $r \geq 0$. The endpoint positions are retained at every iteration.

Theorem 1. *The absolute coefficient sequence of the chromatic polynomial of C_n is infinitely log-concave if and only if $3 \leq n \leq 11$. For $n = 12$ its fifth iterate has a negative entry at degree two; for $13 \leq n \leq 16$ its fourth iterate does; and for every $n \geq 17$ its third iterate does.*

In particular this family refutes Conjecture 21 of Amdeberhan and Moll [1], which proposes infinite log-concavity of the absolute coefficient sequences of all chromatic polynomials. The same proposal appears as Conjecture 13.1 of Amdeberhan [2]. We use the explicit zero-extension convention of Brändén [3]; we do not silently replace it by deleting endpoints at each iteration. No claim that C_{12} is a smallest counterexample among all finite graphs is made.

2 The actual graph and its coefficients

Deleting an edge of C_n produces a path on n vertices, and contracting it produces C_{n-1} when $n \geq 4$. Deletion-contraction and the path coloring count give

$$\chi_n(q) = q(q-1)^{n-1} - \chi_{n-1}(q), \quad \chi_3(q) = q(q-1)(q-2).$$

Induction proves

$$\chi_n(q) = (q-1)^n + (-1)^n(q-1). \quad (1)$$

Consequently the full absolute coefficients, in ascending degree, are

$$a_0 = 0, \quad a_1 = n-1, \quad a_k = \binom{n}{k} \quad (2 \leq k \leq n). \quad (2)$$

The formal proof also verifies the genuine graph correspondence: it constructs a bijection between proper colorings of Mathlib's `SimpleGraph.cycleGraph n` with q colors and closed walks of length n in the complete q -color graph. A trace computation gives (1) for every $q \in \mathbb{N}$, including $q = 0, 1, 2$. Thus the formal sequence is not an assumed binomial surrogate for the graph's chromatic polynomial.

3 A uniform third-iterate obstruction

Put $b = \mathcal{L}a$ and $c = \mathcal{L}b$. For $n \geq 5$, substitution of (2) gives

$$\begin{aligned} b_1 &= (n-1)^2, & b_2 &= \frac{n(n-1)^2(n+4)}{12}, \\ b_3 &= \frac{n^2(n-2)(n-1)^2(n+1)}{144}, & b_4 &= \frac{n^2(n-3)(n-2)^2(n-1)^2(n+1)}{2880}. \end{aligned}$$

A second subtraction gives

$$\begin{aligned} c_1 &= (n-1)^4, \\ c_2 &= \frac{n^2(n-1)^4(n+2)}{16}, \\ c_3 &= \frac{n^3(n-2)^2(n-1)^4(n+1)(n^2+n+18)}{51840}. \end{aligned}$$

These equalities follow by replacing each binomial coefficient by its product formula and collecting factors; all denominators are fixed positive integers. Subtracting c_1c_3 from c_2^2 yields the exact identity

$$(\mathcal{L}^3a)_2 = -\frac{n^3(n-1)^8(n+4)}{103680}(2n^4 - 12n^3 - 327n^2 - 412n + 36). \quad (3)$$

With $n = u + 17$, the final factor becomes

$$2u^4 + 124u^3 + 2529u^2 + 17370u + 6615.$$

It is strictly positive for every $u \geq 0$. All other factors in (3), apart from its minus sign, are positive for $n \geq 17$. This proves the entire infinite negative family in Theorem 1 by one identity and a uniform sign argument.

For example, C_{17} starts with $(a_0, \dots, a_5) = (0, 16, 136, 680, 2380, 6188)$ and $(b_0, \dots, b_4) = (0, 256, 7616, 138720, 1456560)$. Then $c_1 = 65536$, $c_2 = 22491136$, $c_3 = 8150077440$, and

$$(\mathcal{L}^3a)_2 = 22491136^2 - 65536 \cdot 8150077440 = -28272276537344.$$

4 The remaining negative cases

Starting from (2), exact repeated integer subtraction gives

n	iterate r	$(\mathcal{L}^r a)_2$
12	5	−249621701601023742801969101519201265582387397744
13	4	−3618341131654935620812800
14	4	−199158562975246657489530096
15	4	−2734560032125157358883149375
16	4	−25982668618402950000000000000

The particularly large entry for $n = 12$ can be checked from its three preceding neighbors:

$$(\mathcal{L}^4 a)_1 = 45949729863572161, \quad (\mathcal{L}^4 a)_2 = 1505318939996586804452286, \\ (\mathcal{L}^4 a)_3 = 54746933663864342589859620161140.$$

The square of the middle number minus the product of the outer two is the displayed fifth-iterate value. The companion certificate contains the complete preceding sequences for every case, and the Lean declarations `cycle12_negative` through `cycle16_negative` check the same integer statements by kernel-checked finite arithmetic.

5 An invariant proving infinitely many positive iterates

Lemma 2. *Suppose a nonnegative zero-extended finite sequence v satisfies $v_k^2 \geq 3v_{k-1}v_{k+1}$ at every index. Then $w = \mathcal{L}v$ is nonnegative and satisfies the same inequalities.*

Proof. For every index,

$$\frac{2}{3}v_k^2 \leq w_k \leq v_k^2.$$

Thus w is nonnegative, and

$$w_{k-1}w_{k+1} \leq v_{k-1}^2 v_{k+1}^2 \leq \frac{1}{9}v_k^4, \quad w_k^2 \geq \frac{4}{9}v_k^4.$$

It follows that $w_k^2 \geq 4w_{k-1}w_{k+1} \geq 3w_{k-1}w_{k+1}$. Zeros at the boundaries cause no division or exceptional case. $\square$

If finitely many nonnegative initial iterates reach this invariant, induction using Lemma 2 proves nonnegativity of every subsequent iterate. For each $3 \leq n \leq 11$ the following complete finite certificate specifies such an iterate j . Its third column is the minimum of $v_k^2 - 3v_{k-1}v_{k+1}$ over $2 \leq k \leq n-1$, for $v = \mathcal{L}^j a$.

n	j	minimum exact slack
3	0	3
4	1	28
5	2	25825
6	2	90846
7	2	271656
8	3	711608924160
9	3	4029124698225
10	3	19225238518750
11	3	79738750726206

Each row is a directly specified finite integer computation: form (2), iterate the displayed local rule j times, check every preceding entry is nonnegative, then check the listed minimum. Full intermediate arrays are supplied in `exact_certificates.json`. At index one the slack is v_1^2 because $v_0 = 0$; at index zero and the right boundary the assertions are automatic. All indices outside the support remain zero under iteration. This verifies every index, and the invariant proves the positive half of Theorem 1. Together with the negative certificates this completes its proof.

6 Exact reproduction and limits

Run `python3 verify_cycle_counterexample.py` from this paper’s directory. The standard-library checker recomputes the binomial arrays, all finite certificates, deletion–contraction polynomial identities and (3) over exact rational polynomials. Finite calculation proves only the finitely specified certificates; the infinite claims use the invariant and the shifted positive polynomial above, both fully formalized. The result classifies cycles, not all finite graphs or all chromatic coefficient sequences. The earlier public C_{17} counterexample and written classification are retained and credited as predecessors of this full graph-family formalization.

7 Formal verification and reproduction

The principal theorem has a complete Lean proof. The precise scope is: Every genuine cycle graph C_n , $n \geq 3$; full absolute coefficients, zero extension; infinitely log-concave iff $n \leq 11$. The graph/polynomial correspondence and all positive and negative cases are formalized. The public proof source snapshot is fixed at commit `02af7db99db789fecb0d5f5f8f37b1a4e40762a0` in <https://github.com/mxym/math>. The proof packages, with their own dependency locks, audit records and reproduction instructions, are:

- `formalizations/chromatic-cycles-all-n-lean`.

The main formal entry points are:

- `ChromaticCycleAll.cycleGraph_chromatic_polynomial_classification`.
- `ChromaticCycleAll.every_actual_cycle_n_ge_17_fails`.

Lean is fixed to version 4.34.1, compiler commit `5045d0056413266e57c625dcd7c365b10e377c52`; Mathlib is fixed to `d13f23b723b8a846827a245b89c10fc7d3f11612`. The audited mathematical roots depend only on the standard axioms `propext`, `Classical.choice`, and `Quot.sound`. The archived audits state their actual compilation and replay scopes; publication of this typeset version does not imply a new execution of every previous empty-kernel replay. The companion README maps each printed principal result to its formal theorem and explains how to repeat verification.

Research and preparation disclosures

This research, formalization and manuscript preparation used AI assistance. The author is responsible for the manuscript. The principal claims are supported by the readable proofs and the public Lean artifacts specified above. Formal verification is not external human peer review; no such review is claimed. No external funding was received. No worldwide priority or journal acceptance is asserted.

References

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- [3] P. Brändén, *Iterated sequences and the geometry of zeros*, arXiv:0909.1927, especially the zero-extension convention. <https://arxiv.org/abs/0909.1927>.